

Hands-On Lab: Generative AI for Data Preparation

Estimated duration: 30 minutes

Overview

In this lab, you will learn how to use generative AI to prepare data using the tool ChatGPT.

Objective(s)

After completing this lab, you will be able to:

- Sign in on <https://chat.openai.com/>
- Upload a dataset
- Handle missing values
- Perform data standardization
- Perform data normalization

Pre-requisite(s)

- A free ChatGPT account
- A basic understanding of **Exploratory Data Analysis (EDA)**

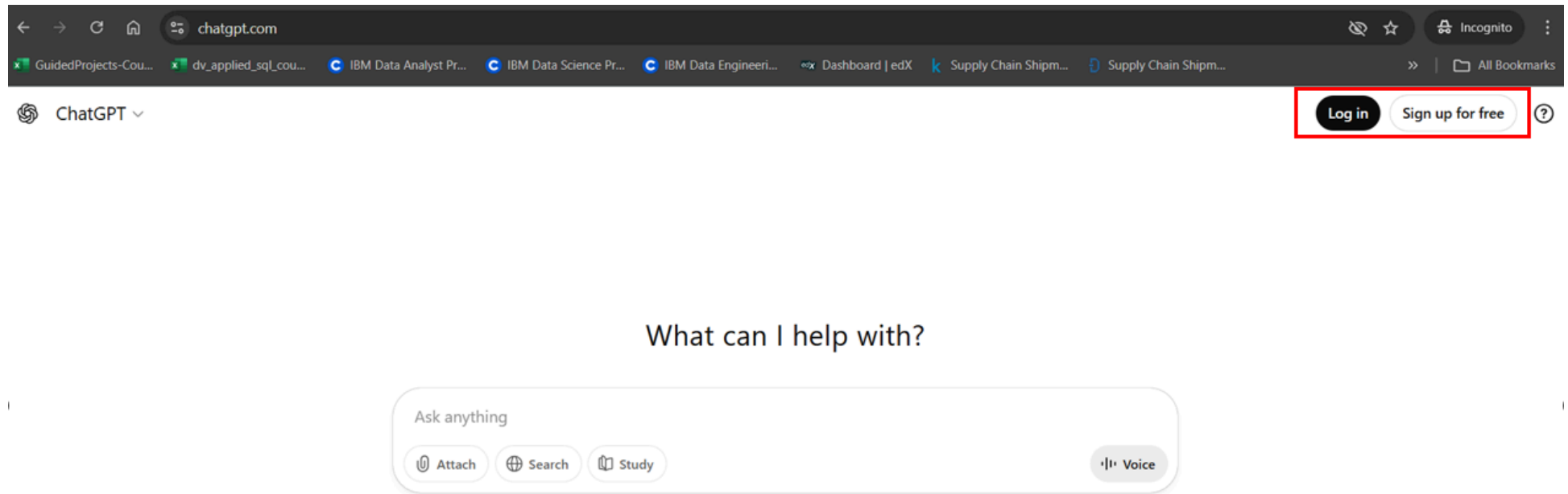
Dataset

The dataset is a **filtered and modified version** of the [Laptop Price Prediction using specifications dataset](#), available under the **Database Contents License (DbCL) v1.0 on the Kaggle website**.

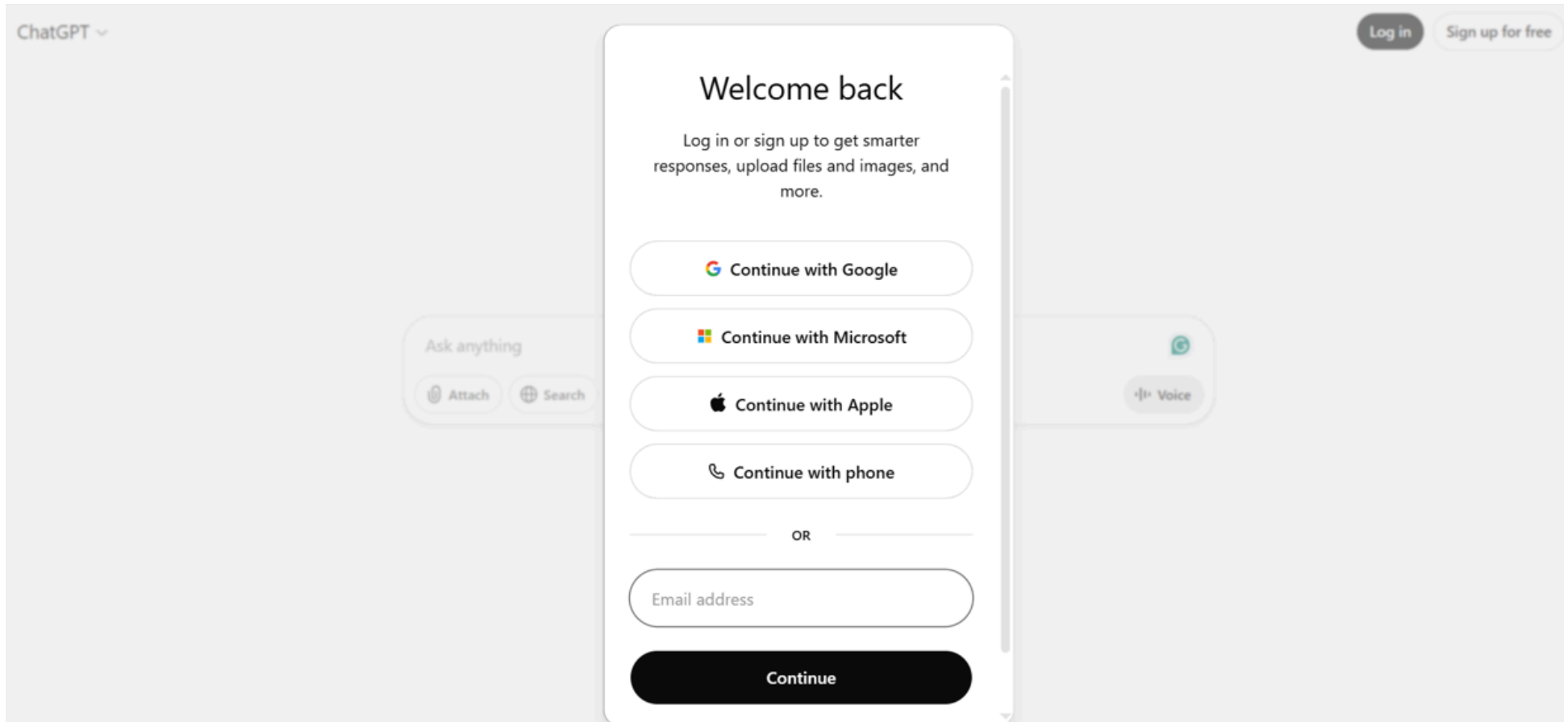
While holding down the **Ctrl (Windows) or Command (Mac) button**, click [here](#) to download the data set.

Task 1: Sign in on ChatGPT

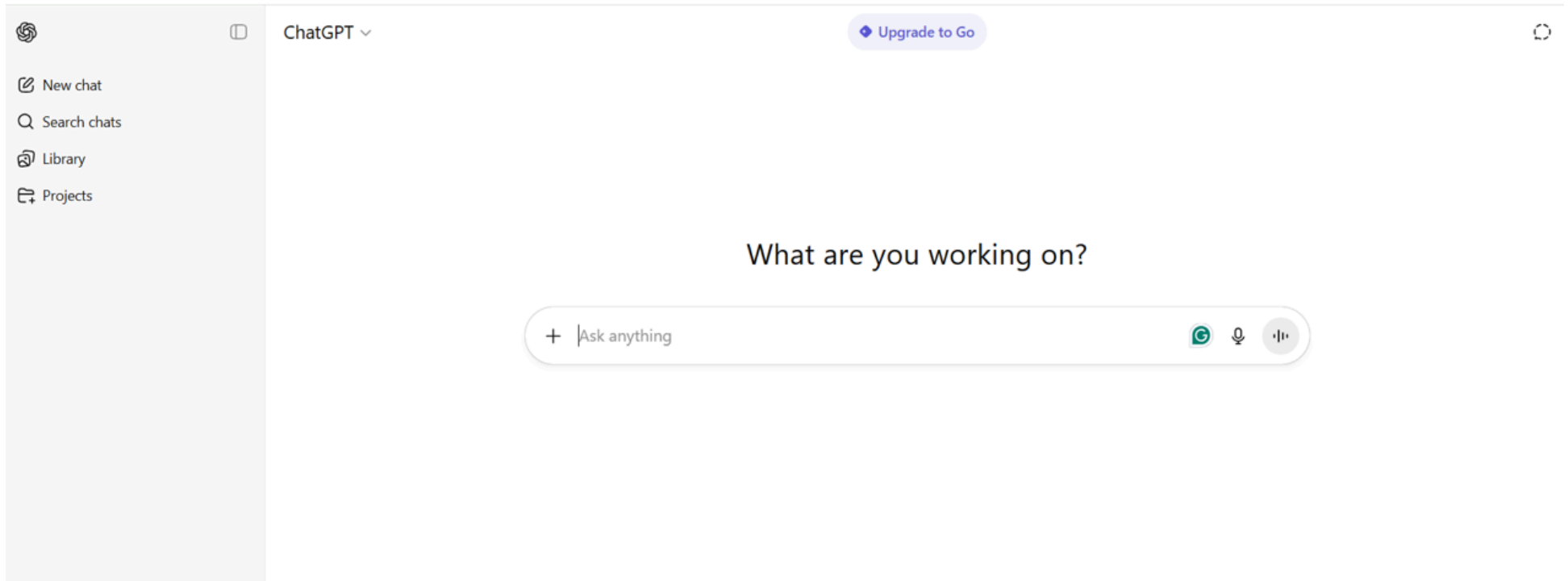
Step 1: If you do not have an account, go to <https://chat.openai.com/>, and click **Log in or Sign up for free** at the top right corner.



Step 2: Log in or sign up using any of the options — Google, Microsoft, Apple, GitHub, or Email — and follow the prompts to sign in.

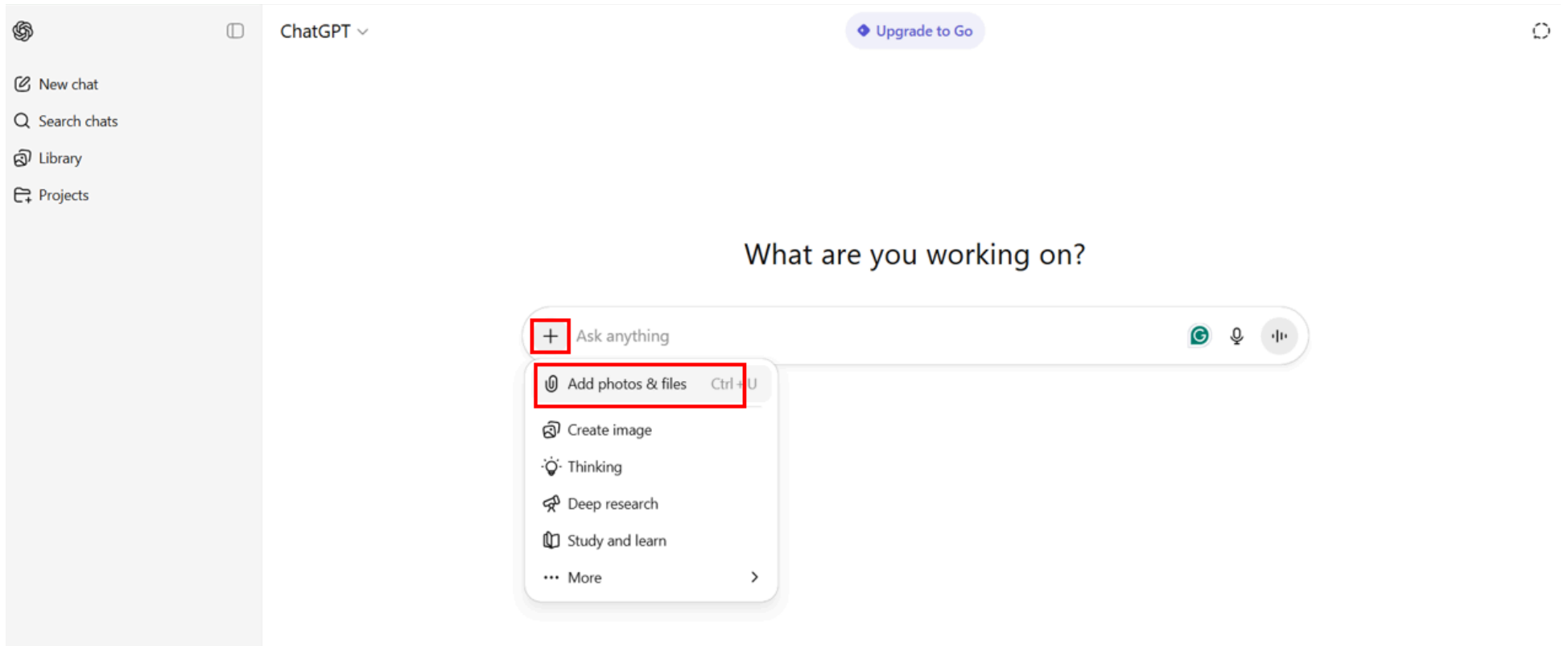


Step 3: After creating your account, log in with your credentials, and the ChatGPT interface will be displayed.



Task 2: Upload Dataset

Step 1: On the **ChatGPT** interface, click the **Add (+)** icon and choose **Add photos & files** to upload the Laptop Pricing Data that you have downloaded.



Step 2: Once the file is uploaded, write a prompt **Describe the dataset** to get the statistical description of the dataset. Copy the prompt below, paste it into the ChatGPT interface, and press Enter.

1	Describe the dataset	 
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The screenshot shows the Skills Network Editor interface. On the left is a sidebar with a ChatGPT logo and a menu containing 'New chat', 'Search chats', 'Library', and 'Projects'. The main area is titled 'ChatGPT' with a dropdown arrow. In the top right of the main area, there is a blue button that says 'Upgrade to Go'. Below this, the text 'What are you working on?' is displayed. A file upload box contains a green spreadsheet icon, the filename 'laptop_pricing_dataset_mod1..csv', and the word 'Spreadsheet'. Below the filename is a plus sign and the text 'Describe the dataset'. To the right of this text are three icons: a green circle, a microphone, and a black circle with a white upward arrow.

Step 3: Scroll down through the dataset output to view the details presented by ChatGPT.

ChatGPT ▾

 Sharelaptop_pricing_dataset_mod1..csv
Spreadsheet

Describe the dataset

Here's a summary of the **laptop pricing dataset** you uploaded:

- **Number of rows:** 238
- **Number of columns:** 13

Column Overview

Column	Type	Missing Values	Description (inferred)
Unnamed: 0	int64	0	Likely an index column.
Manufacturer	object	0	Brand name (e.g., Acer, Dell, HP).
Category	int64	0	Encoded laptop type or segment.
Screen	object	0	Type of screen (e.g., IPS Panel, Full HD).
GPU	int64	0 	Encoded GPU type or manufacturer.

 Ask anything

Task 3: Handle missing values

Step 1: To identify the missing values, write a prompt:

1 ****Identify the attributes with missing data****

and press **Enter**.

The response will display two attributes with missing values as:

- **"Screen_Size_cm"** - categorical variable
- **"Weight_kg"** - continuous variable.

To replace the missing values, copy the prompt below, paste it into the ChatGPT interface, and press Enter.

```
1 - Missing entries in columns containing categorical values need to be replaced with the most frequent entries
2 - Missing entries in columns with continuous data need to be replaced with the mean value of the column. If a value is
```


The prompt response will be shown as below:

ChatGPT 

 Share

Identify the attributes with missing data

The dataset contains **two attributes with missing data**:

Attribute	Missing Values	Percentage of Total
Screen_Size_cm	4	1.68%
Weight_kg	5	2.10%

 All other columns are **complete (no missing values)**.

Would you like me to show how these missing values are distributed (e.g., which manufacturers or categories they belong to)?

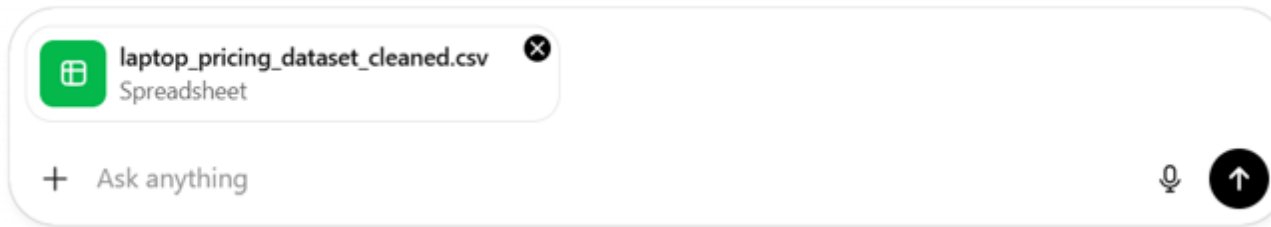


+ Ask anything



Step 2: Write a prompt:

1 Replace the missing values with the mean value in the respective column and save the updated dataset.



Step 2: Write the following prompts to change the attributes into standard units:

- 1 - Data under Screen_Size_cm; is assumed to be in centimeters. Convert this data into inches. Modify the name of the attribute to Screen_Size_inches.
- 2 - Data under Weight_kg is assumed to be in kilograms. Convert this data into pounds. Modify the name of the attribute to Weight_pounds.
- 3 - Provide the updated dataset as a downloadable CSV.

ChatGPT ▾

 Share

The missing values in `Screen_Size_cm` and `Weight_kg` have been replaced with their respective mean values.

You can download the cleaned dataset using this link:

 [laptop_pricing_dataset_cleaned.csv](#) 

     ...



laptop_pricing_dataset_cleaned.csv 
Spreadsheet

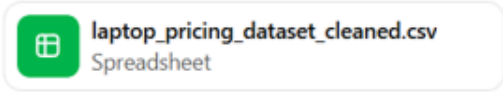
Convert the 'Screen_Size_cm' values from centimeters to inches and rename the column to 'Screen_Size_inch'.
Convert the 'Weight_kg' values from kilograms to pounds and rename the column to 'Weight_pounds'.
Provide the updated dataset for download as a CSV.



ChatGPT will generate and **provide the new dataset link**. You can **download the CSV** and open it in Excel if needed.

ChatGPT ▾ Share



Convert the 'Screen_Size_cm' values from centimeters to inches and rename the column to 'Screen_Size_inch'.
Convert the 'Weight_kg' values from kilograms to pounds and rename the column to 'Weight_pounds'.
Provide the updated dataset for download as a CSV.

The conversions and renaming have been completed successfully.

You can download the updated dataset here:

+ Ask anything

Task 5: Perform the data normalization

Some attributes might require normalization. Since there are multiple normalization methods, you need to specify your exact requirements and tasks in the prompt.

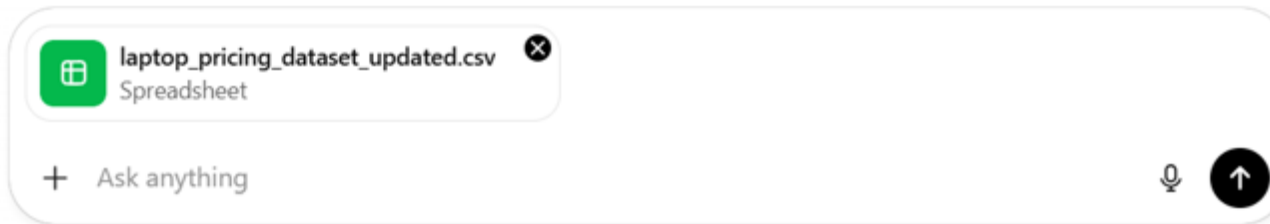
You can either:

- Save the normalized data as a new attribute, or
- Overwrite the original attribute with the normalized values.

Example: Assume the data under 'CPU_frequency' needs to be normalized using the maximum value of the attribute. You want the normalized values to replace the original attribute instead of creating a new one.

Make sure to provide clear instructions in your prompt so ChatGPT can process the normalization correctly.

Step 1: Upload the new updated dataset from the previous step by clicking the '+' icon and attach the new dataset.



Step 2: Write the following prompt:

- 1 - Normalize the values under the column 'CPU_frequency' with respect to its maximum value.
- 2 - Do not create a new attribute; replace the existing values.
- 3 - Provide the updated dataset as downloadable CSV.

ChatGPT ▾

[Upgrade to Go](#) ×[Share](#)**laptop_pricing_dataset_updated.csv**
Spreadsheet

Normalize the values under the column 'CPU_frequency' with respect to its maximum value.
Do not create a new attribute; replace the existing values.
Provide the updated dataset as downloadable CSV.

Here's your updated dataset with the 'CPU_frequency' column normalized (values replaced in place):

[Download laptop_pricing_dataset_normalized.csv](#)



+ Ask anything



Practice problems

- Create a prompt to generate a Python code that converts the values under Price from USD to Euros.
- Modify the normalization prompt to perform min-max normalization on the CPU_frequency parameter.

Conclusion

In this lab, you have learned to handle missing values in your dataset, and performed data standardization and data normalization.

Author(s)

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